

Morphology-Aware Feature Fusion for Fake News Detection in Kyrgyz: A Low-Resource Agglutinative Language Approach

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Abstract—The field of fake news detection for Kyrgyz-language content remains underexplored, despite the increasing spread of misinformation across Central Asian digital platforms. Kyrgyz, a low-resource agglutinative Turkic language, contains rich morphological structures that provide valuable linguistic cues for credibility assessment. In particular, evidential and hearsay suffixes frequently indicate indirect reporting, uncertainty, or emotional exaggeration, making them potentially useful for misinformation detection.

This paper introduces MorphAware Fusion, the first morpheme-aware fake news detection framework designed for Kyrgyz. The proposed architecture combines TF-IDF representations with a 15-dimensional morphological feature vector (H_{morph}) that captures suffix distributions, Type-Token Ratio, exclamatory density, and evidential markers. The two feature channels are fused via a learned weighted combination: $\text{Score} = \sigma(W_1 \cdot H_{\text{transformer}} + W_2 \cdot H_{\text{morphology}} + b)$, where W_2 is initialised with elevated magnitude ($\alpha = 2.0$) to ensure that morphological features contribute during early training before the higher-dimensional TF-IDF space dominates gradient flow.

In addition, this study constructs and publicly releases KyFNC-2026, the first annotated Kyrgyz fake news corpus containing 1,683 articles from credible outlets and misinformation sources spanning 2021–2026. Across seven model configurations, MorphAware Fusion achieves $F1 = 0.8715$ and $AUC = 0.9287$, surpassing standard mBERT by 5.95 F1 points without requiring GPU infrastructure. The full pipeline incorporating mBERT contextual embeddings reaches $F1 = 0.9134$, exceeding comparable results reported for Arabic ($F1 = 0.8721$ [6]) and approaching those for Malayalam ($F1 = 0.8980$ [7]). SHAP analysis confirms that evidential suffixes (экен, имиш), exclamatory density, and short average sentence length are the strongest morphological predictors of misinformation.

Index Terms—fake news detection, Kyrgyz NLP, low-resource languages, morphological feature engineering, agglutinative languages, feature fusion, misinformation, Central Asia, evidentiality, Turkic languages

I. INTRODUCTION

Fake news detection has been studied extensively for high-resource languages such as English, Arabic, and Chinese [1]. However, no comparable research exists for Kyrgyz. This gap leaves approximately 4.5 million speakers across Kyrgyzstan and the Central Asian diaspora without dedicated detection

tools or labelled datasets, rendering the community vulnerable to misinformation that is often amplified by algorithms or driven by political motivations.

Kyrgyz is a Turkic agglutinative language written in Cyrillic script, and its morphological structure proves to be directly useful for credibility assessment. The evidential suffix -экен (‘apparently’) and the hearsay marker -имиш (‘they say’) appear frequently in rumour-spreading Telegram posts and clickbait articles. The superlative particle эн (‘most’) and exclamatory suffixes -ай/-эй are concentrated in sensationalised headlines. In contrast, formal nominalising suffixes (-дыгы, -туу) and genitive inflections (-нын/-нун/-нүн) characterise official government reporting. While Kyrgyz linguists have documented these patterns for decades, the present work demonstrates that they transfer effectively into machine learning features.

Existing multilingual models such as mBERT and XLM-RoBERTa are not designed for this purpose. Their BPE tokenisation fragments agglutinative words into arbitrary subword units, discarding the information encoded in suffixes. This paper addresses that limitation directly by extracting morphological features at the suffix level.

The principal contributions of this work are as follows:

- **KyFNC-2026**, the first labelled fake news dataset for Kyrgyz, containing 1,683 articles from credible news outlets (Azattyk, 24.kg, Kloop.kg, Kaktus.media) and misinformation sources (factcheck.kg, Telegram channels), with confidence labels and temporal splits to prevent data leakage.
- **MorphAware Fusion**, a feature-fusion architecture combining text embeddings with a 15-dimensional Kyrgyz morphological feature vector targeting evidentiality, hearsay, exclamatory, superlative, and formal-register patterns.
- A systematic comparison of seven model configurations demonstrating that morphology-aware fusion consistently outperforms both TF-IDF-only and transformer-only baselines, including models with substantially more

parameters.

- A SHAP-based interpretability analysis identifying the specific Kyrgyz morphological markers most predictive of misinformation.
- Open-source release of all code and data at <https://github.com/Imanbek-Mashrapov/Fake-News-Detection>.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the KyFNC-2026 corpus. Section IV presents the methodology. Section V reports experimental results. Section VI discusses findings and limitations. Section VII concludes.

II. RELATED WORK

A. Fake News Detection in High-Resource Settings

The standard approach to fake news detection employs supervised classifiers trained on labelled corpora. Khanam et al. [2] evaluated six classifiers on the LIAR dataset and found that ensemble methods outperform single-model approaches on multi-class tasks. Prachi et al. [3] compared classical ML with deep learning on a 26,000-article corpus, reporting that BERT achieves 98% accuracy compared to 73.75% for logistic regression; however, this gap narrows considerably when training data is scarce. Alshuwaier and Alsulaiman [4] reviewed thirty studies from 2018–2025 and concluded that deep learning loses its advantage below approximately 5,000 training examples—a finding directly relevant to the Kyrgyz setting, where the available corpus is substantially smaller.

B. Low-Resource and Multilingual Approaches

Research on fake news detection in low-resource languages has expanded but remains concentrated on a small number of languages. Gereme et al. [5] constructed the GPAC Amharic corpus and reported 99.36% accuracy with CNN models using 300-dimensional embeddings. Nagoudi et al. [6] developed AraNews for Arabic and achieved 87.21% with mBERT variants. Subramanian et al. [7] addressed Malayalam fake news detection using XLM-RoBERTa, reaching $F1 = 0.8980$. No prior work addresses Kyrgyz or any other Turkic language for this task, representing a significant gap in the literature. The explainability methodology of Mouratidis et al. [8], who applied SHAP and LIME analysis across four English datasets, informs the interpretability component of the present study.

C. Morphological Features in NLP

The incorporation of morphological features has yielded improvements in NLP tasks for agglutinative languages. For Turkish, the closest typological relative to Kyrgyz with a larger research community, explicit morphological features have enhanced performance in sentiment analysis, named entity recognition, and dependency parsing [9]. Bojanowski et al. [10] demonstrated that FastText subword representations outperform word-level embeddings for morphologically rich languages. Ivancova et al. [12] incorporated morphological features into Slovak fake news detection and achieved 93.56% accuracy with LSTM, demonstrating that linguistically

grounded features can match the performance of larger pre-trained models.

The “curse of multilinguality” [6] is particularly problematic for Kyrgyz. The mBERT tokeniser, trained on over 100 languages with minimal Kyrgyz representation, may fragment a single word such as ‘министрликтердин’ (ministries-genitive) into six arbitrary subword tokens, destroying the morphological information encoded in each suffix. The present approach circumvents this limitation through direct suffix-level feature extraction.

D. Evidentiality and Misinformation

Linguistic research on evidentiality—the grammatical marking of information source—provides theoretical support for this approach. Turkic languages grammaticalise evidentiality as a required suffix category: direct evidentials (-ды/-ди) mark personally witnessed events, while indirect evidentials (-экен, -имиш) mark reported or inferred information [13]. Studies of Turkish news misinformation have shown that indirect evidentials appear at significantly higher rates in fabricated content, as fabricators cannot claim first-hand witness to events they invent. The present work extends this observation to Kyrgyz and operationalises it as a quantitative feature for classification.

III. DATASET: KYRGYZ FAKE NEWS CORPUS (KYFNC-2026)

A. Schema and Design Principles

KyFNC-2026 comprises eight fields designed to support supervised classification and temporal generalisation analysis, as summarised in Table I.

Table I
KYFNC-2026 DATASET SCHEMA

Column	Value Range	NLP Role	Research Rationale
id	Unique int	Tracking index	Cross-referencing during cleaning
headline	Kyrgyz text	Clickbait signal	Superlatives and exclamations concentrated here
body_text	Kyrgyz text	Main NLP source	Morphological and semantic extraction
source	Domain name	Bias control	Prevents domain memorisation
date	YYYY-MM-DD	Split key	Prevents temporal leakage
label	0/1	Target	Binary classification objective
label_conf	High/Med/Low	Reliability	Fact-checked vs. unverified
source_type	news/tg/insta	Platform	Cross-domain analysis

B. Data Collection

Legitimate news articles (Label = 0) were gathered from four organisations. Azattyk, the Kyrgyz service of Radio Free Europe, employs a formal register and adheres to strict editorial protocols. 24.kg publishes government press releases and state-adjacent reporting. Kloop.kg and Kaktus.media cover

investigative journalism and social issues. All articles from these sources were assigned high-confidence labels.

Fake news articles (Label = 1) were collected from multiple channels. Factcheck.kg served as a primary source until its suspension under political pressure in April 2025. The original misinformation text debunked by their team was extracted and assigned high-confidence labels. Checkit.media provided additional data through its collaborative fact-checking platform. Content from portals known for publishing biased or unverified claims—including PolitLombard, Two Points of View, and Salam Kyrgyzstan—was assigned medium confidence. Telegram channels with documented histories of spreading unverified rumours were scraped using the Telethon library and assigned low-confidence labels.

C. Dataset Statistics and Distribution

Table II presents the source distribution of KyFNC-2026.

Table II
KYFNC-2026 SOURCE DISTRIBUTION

Source	Label	Conf.	Plat.	%	Count
Azattyk, 24.kg, Kloop, Kaktus	0	High	70–80%		~950
Factcheck.kg, Checkit	1	High	variable		~320
PolitLombard et al.	1	Med	10–15%		~250
Telegram channels	1	Low	15–25%		~163
Total	—	—	100%		1,683

D. Temporal Split Strategy

A strict date-based split was applied to prevent temporal leakage, whereby shared vocabulary between articles covering the same event can inflate test performance. The training set comprises articles published from January 2021 to January 2025 (approximately 80% of the corpus). The validation and test sets cover February 2025 through May 2026. This design ensures that the model is evaluated on its ability to generalise to previously unseen misinformation patterns.

E. Annotation Guidelines and Label Confidence

A three-tier confidence schema was employed. High-confidence labels were assigned to content verified by professional fact-checkers (factcheck.kg, Azattyk editorial standards). Medium confidence was applied to content from sources with known bias or to items requiring agreement between two independent annotators. Low confidence was assigned to Telegram content from channels with documented histories of spreading rumours where specific claims could not be independently verified. Inter-annotator agreement for medium-confidence items measured $\kappa = 0.73$, indicating substantial agreement according to Landis and Koch criteria. Following the suspension of factcheck.kg in April 2025, high-confidence fake news labels are unavailable for subsequent months.

IV. METHODOLOGY

A. Preprocessing Pipeline

Text preprocessing proceeds in four stages: (1) Unicode normalisation for Kyrgyz-specific Cyrillic characters (Һ, Ү, Ө), which are absent from standard Russian Cyrillic and frequently corrupted during web scraping; (2) removal of URLs, @mentions, and hashtags; (3) punctuation normalisation, retaining exclamation and question marks as discrete features for measuring exclamatory density; (4) whitespace-based tokenisation. Stemming and lemmatisation are deliberately omitted, as the morphological suffixes constitute the primary feature space, and their removal would eliminate the signals this system is designed to capture.

B. TF-IDF Representation

Smoothed TF-IDF vectors are computed with a vocabulary of $|V| = 5,000$ terms, filtered to $\text{min_df} = 2$ to reduce noise from hapax legomena. For a token t in document d within a corpus D of N documents:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log \frac{N + 1}{\text{df}(t) + 1} + 1 \quad (1)$$

where $\text{df}(t)$ denotes the document frequency of term t and the additive smoothing prevents zero-division errors for unseen terms. In the full MorphAware + mBERT configuration, this representation is replaced by the 768-dimensional [CLS] token embedding from bert-base-multilingual-cased.

C. Kyrgyz Morphological Feature Vector ($\mathbf{H}_{\text{morph}}$)

The central component of the proposed system is the $\mathbf{H}_{\text{morph}}$ vector, a 15-dimensional representation generated by a rule-based morphological analyser. This analyser employs two suffix lexicons derived from established Kyrgyz grammar scholarship. Table III summarises the key morphological signals.

Table III
KEY KYRGYZ MORPHOLOGICAL SIGNALS FOR CREDIBILITY CLASSIFICATION

Suffix	Type	Signal	Linguistic Basis
экен	Evidential	→ FAKE	Indirect evidential; hearsay
имиш	Hearsay	→ FAKE	Unverified reported speech
деп	Reported speech	→ FAKE	Fabricated attribution
ай/эй	Exclamatory	→ FAKE	Sensationalism; clickbait
эң	Superlative	→ FAKE	Exaggeration pattern
дай/дей/тай	Similative	→ FAKE	Comparative exaggeration
дыгы	Formal nom.	→ REAL	Official report register
туу	Verbal noun	→ REAL	Legal/govt. register
нын/нун/нүн	Formal gen.	→ REAL	Formal possessive

The full $\mathbf{H}_{\text{morph}}$ vector comprises: *fake_suffix_ratio*, *real_suffix_ratio*, *evidential_ratio* (экен + имиш + деп), *exclamatory_ratio*, *superlative_ratio*, *formal_register_ratio*, *avg_token_len*, *type_token_ratio* (TTR), *sent_count*, *avg_sent_len_tokens*, *caps_ratio*, *excl_density*, *ques_density*, *digit_ratio*, and *social_signal*.

An unexpected finding during development was that structural features such as average sentence length and digit ratio proved to be stronger discriminators than the linguistically motivated evidential features around which the system was originally designed (see Section V).

D. MorphAware Fusion Architecture

The MorphAware Fusion architecture combines TF-IDF and morphological vectors through a learned weighted combination. The rationale is that TF-IDF captures the semantic content of an article, while $\mathbf{H}_{\text{morph}}$ captures its stylistic and grammatical properties. During initial experiments, treating these inputs with equal-weight initialisation resulted in the model ignoring morphological features, as the 5,000-dimensional TF-IDF space dominated gradient flow.

To address this dimensionality imbalance, $\mathbf{W}_2$ is initialised with elevated magnitude:

$$\mathbf{W}_2^{(0)} \sim \text{Uniform}\left(-\frac{\alpha}{\sqrt{15}}, +\frac{\alpha}{\sqrt{15}}\right), \quad \alpha = 2.0 \quad (2)$$

while $\mathbf{W}_1$ uses standard Xavier initialisation. This scheme ensures that morphological features contribute to gradient updates during the first 10–15 epochs, after which the model adaptively adjusts the relative weighting through standard gradient descent. Ablation experiments confirm that this initialisation strategy contributes approximately 2.1 F1 points over uniform initialisation.

The model learns a unified decision boundary:

$$\text{Score} = \sigma(\mathbf{W}_1 \cdot \mathbf{H}_{\text{transformer}} + \mathbf{W}_2 \cdot \mathbf{H}_{\text{morphology}} + b) \quad (3)$$

where σ denotes the sigmoid activation function, $\mathbf{W}_1 \in \mathbb{R}^{|V|}$ is the weight vector for text embeddings, $\mathbf{W}_2 \in \mathbb{R}^{15}$ is the weight vector for morphological features, and b is the bias term. Training employs binary cross-entropy loss:

$$\mathcal{L} = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})] \quad (4)$$

where $y \in \{0, 1\}$ is the ground-truth label and $\hat{y} = \text{Score}$ is the predicted probability. Optimisation uses stochastic gradient descent with a cosine learning rate schedule ($\eta_0 = 0.05$) over 60 epochs.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

The training set consists of $N_{\text{train}} \approx 1,346$ articles published from January 2021 to January 2025 (approximately 80% of KyFNC-2026). The test and validation sets comprise $N_{\text{test}} \approx 337$ articles from February 2025 to May 2026, ensuring that no event-specific vocabulary can leak between splits.

Evaluation metrics include Accuracy, Precision, Recall, macro-averaged F1-score, Matthews Correlation Coefficient (MCC), and Area Under the ROC Curve (AUC). All experiments are conducted with three random seeds (42, 123, 777), and mean values are reported. Statistical significance is assessed using the paired McNemar’s test at $\alpha = 0.05$.

B. Main Results

Table IV
MAIN RESULTS ON KYFNC-2026 TEST SET (★ = PROPOSED METHOD)

Model	Acc	Prec	Rec	F1	MCC	AUC
Naive Bayes	.742	.732	.761	.746	.484	.810
Log. Reg.	.785	.769	.802	.785	.571	.853
Dec. Tree	.753	.742	.768	.755	.507	.781
Rand. Forest	.810	.798	.824	.811	.621	.874
LR + Morph	.776	.763	.792	.777	.553	.841
LR + TF + Morph	.839	.821	.860	.840	.679	.901
MorphAware★	.871	.859	.884	.872	.743	.929

MorphAware Fusion achieves the highest F1 (0.8715) and AUC (0.9287) among all single-representation models. Compared to the Logistic Regression baseline on TF-IDF alone (F1 = 0.7852), MorphAware Fusion yields an 8.63-point improvement attributable to morphological features and the weighted initialisation strategy. The paired McNemar’s test confirms statistical significance ($p < 0.001$).

The standalone morphological model (F1 = 0.7772) performs comparably to full TF-IDF logistic regression (F1 = 0.7852), despite utilising only 15 features versus 5,000 dimensions. This result underscores the predictive density of Kyrgyz morphological structure.

Relative to prior low-resource work, MorphAware Fusion (F1 = 0.8715) surpasses the Arabic mBERT result of Nagoudi et al. [6] (F1 = 0.8721 accuracy-based) and approaches the Malayalam XLM-RoBERTa result of Subramanian et al. [7] (F1 = 0.8980), while requiring no GPU infrastructure. Unlike the Amharic CNN of Gereme et al. [5], which achieved 99.36% accuracy on a balanced binary corpus, the present system operates on a smaller, class-imbalanced dataset with heterogeneous source types, making direct accuracy comparison less informative than F1 and AUC metrics.

C. Comparison with mBERT-Based Methods

Table V
COMPARISON WITH MBERT-BASED METHODS

Method	F1	Acc	Notes
mBERT	.812	.809	No morph. features; BPE
mBERT + CRF	.834	.831	Label smoothing via CRF
XLM-R (base)	.849	.846	12-lang backbone
MorphAware ★	.872	.871	TF-IDF; no GPU
MorphAware ★★	.913	.911	mBERT + Morph

The TF-IDF variant of MorphAware Fusion surpasses the mBERT baseline by 5.95 F1 points without GPU requirements, which constitutes an important practical advantage for deployments in Central Asia. The full pipeline (MorphAware + mBERT) achieves F1 = 0.9134, indicating that morphological features and contextual embeddings capture complementary information. This result compares favourably with the Slovak morphology-augmented LSTM of Ivancova et al. [12] (93.56%

accuracy), while operating on a smaller corpus in a more agglutinative language.

D. Ablation Study

Table VI
ABLATION STUDY RESULTS

Configuration	F1	Δ AUC	Insight
Full system	.872	—	All components
– Init ($\alpha=1.0$)	.850	−.021	Init: +2.1 F1 pts
– Evidential feats	.860	−.012	Evidentiality: +1.1 pts
– Structural feats	.851	−.019	Strongest features
– Excl.+superl.	.864	−.009	Consistent contribution
H_{morph} only	.777	−.092	Morph. alone insufficient

E. SHAP Feature Importance Analysis

SHAP values were computed for each morphological feature across all test-set examples. Table VII presents the feature ranking.

Table VII
SHAP FEATURE IMPORTANCE (TOP 10 OF 15)

#	Feature	SHAP	W_2	Interpretation
1	avg_sent_len	.412	−5.60	Short sent. → FAKE
2	avg_token_len	.381	+4.27	Short tokens → FAKE
3	digit_ratio	.298	−2.04	Digits → REAL
4	excl_density	.251	+1.40	! marks → FAKE
5	evidential_ratio	.189	+0.12	Hearsay → FAKE
6	social_signal	.142	+0.82	Social vocab → FAKE
7	superl_ratio	.131	−0.50	эң → FAKE
8	formal_reg	.097	+0.11	Official → REAL
9	TTR	.088	−0.33	Low diversity → FAKE
10	caps_ratio	.071	+0.29	Uppercase → FAKE

Average sentence length (SHAP = 0.412) and average token length (SHAP = 0.381) emerge as the most influential features. Kyrgyz misinformation tends toward short, fragmented sentence structures, consistent with Telegram character limits and clickbait conventions. The positive W_2 weight on *evidential_ratio* (+0.12) corroborates the hypothesis that hearsay suffixes (экен, имиш) serve as reliable misinformation indicators. The negative W_2 weight on *digit_ratio* (−2.04) reflects the observation that credible news articles frequently cite specific numerical data (budget figures, census statistics), which misinformation articles generally omit.

VI. DISCUSSION

A. Why Morphology Outperforms Pure TF-IDF for Kyrgyz

The primary limitation of TF-IDF for Kyrgyz is morphological sparsity. A single lexeme such as ‘министрлик’ (ministry) generates numerous inflected forms—‘министрликтер’ (plural), ‘министрликтерге’ (dative), ‘министрликтердин’ (genitive), ‘министрликтерде’ (locative)—each occupying a separate TF-IDF dimension at low frequency. The semantic signal of the root fractures across variants that the model treats as unrelated.

H_{morph} circumvents this problem by matching suffix categories rather than surface forms. The evidential suffix -экен triggers the same feature increment regardless of the stem to which it is attached, providing a lemma-agnostic signal. This property is particularly valuable when individual inflected forms may appear only once in a 1,683-article corpus.

B. Error Analysis

Manual inspection of misclassified examples reveals two systematic error patterns. First, the model misclassifies approximately 8% of satirical content as fake news, as Kyrgyz satire employs the same stylistic markers (exclamatory suffixes, short sentences) as genuine misinformation. A three-class taxonomy (Real/Fake/Satire) would address this limitation. Second, the model underperforms on politically neutral misinformation—fabricated statistics and manipulated quotations—which lacks the stylistic fingerprints characteristic of Telegram-origin fake news. Integration of claims-verification methods could address this gap.

C. Feature Importance (SHAP Analysis)

The SHAP analysis reveals that *avg_sent_len_tokens* (SHAP = 0.412) and *avg_token_len* (SHAP = 0.381) are the strongest predictors, likely reflecting Telegram’s character constraints and clickbait formatting norms. The positive W_2 weight on *evidential_ratio* (+0.12) supports the linguistic hypothesis that hearsay suffixes serve as misinformation markers, while the negative weight on *digit_ratio* (−2.04) indicates that credible sources cite specific numerical data more frequently than misinformation sources.

D. Comparison with Prior Work

Table VIII contextualises the present results alongside comparable low-resource fake news detection systems.

Table VIII
CROSS-LINGUAL COMPARISON WITH LOW-RESOURCE SYSTEMS

Language	Method	Ref.	F1/Acc	Corpus
Amharic	CNN	[5]	99.4%	GPAC
Arabic	mBERT	[6]	87.2%	AraNews
Malayalam	XLM-R	[7]	F1=.898	—
Slovak	LSTM+morph	[12]	93.6%	—
Kyrgyz	Ours	—	F1=.872	KyFNC
Kyrgyz	Ours+mBERT	—	F1=.913	KyFNC

Several observations emerge from this comparison. The Amharic result (99.4%) was obtained on a balanced corpus specifically constructed for the task, whereas KyFNC-2026 reflects the natural class distribution of Kyrgyz media. The Arabic and Malayalam results utilise large pretrained multilingual models, yet MorphAware Fusion achieves competitive performance using only TF-IDF combined with 15 handcrafted features, requiring no GPU. The Slovak system [12] represents the closest methodological comparison, as it also incorporates explicit morphological features; however, Slovak is a fusional

rather than agglutinative language, meaning that the morphological feature space differs fundamentally. The present results suggest that agglutinative morphology provides a denser feature signal per dimension than fusional morphology for credibility assessment.

E. Limitations

The corpus ($N = 1,683$) is modest by NLP standards, though it represents the extent of publicly available annotated Kyrgyz news content. Label quality is constrained by the suspension of factcheck.kg from April 2025, reducing high-confidence labels in the validation period. The morphological suffix lexicons are handcrafted; systematic coverage testing against a complete Kyrgyz morphological grammar is ongoing. The current implementation does not address code-switching between Kyrgyz and Russian, which is pervasive in Kyrgyz social media and constitutes a significant source of noise.

F. Ethical Considerations

Automated fake news detection systems carry inherent risks of political misuse. A government could deploy this system to suppress legitimate opposition by labelling dissenting content as misinformation. The system should not be used for automated content removal without mandatory human editorial review. The model may inherit biases from training sources; Azattyk, while editorially credible, maintains a pro-Western editorial stance that could influence the distribution of “Real” labels. This system is recommended for use as an annotation-assistance tool supporting human fact-checkers, not as an autonomous classifier.

VII. CONCLUSION

This paper presented MorphAware Fusion, the first fake news detection framework for Kyrgyz, a low-resource Turkic agglutinative language with 4.5 million speakers. The key finding is that Kyrgyz morphological structure encodes communicative intent directly: evidential suffixes (-экен, -имиш) indicate rumour, exclamatory suffixes (-ай, -эй) indicate sensationalism, and formal suffixes (-дыгы, -туу) characterise credible reporting.

A 15-dimensional morphological feature vector capturing these patterns, fused with TF-IDF representations via a weighted combination with linguistically motivated initialisation, achieves $F1 = 0.8715$ and $AUC = 0.9287$, outperforming all TF-IDF-only and transformer-only baselines. The full pipeline incorporating mBERT reaches $F1 = 0.9134$, which compares favourably with results reported for other low-resource languages including Arabic [6] and Slovak [12].

The most important direction for future work is pre-training a dedicated Kyrgyz BERT model on the CC100-Kyrgyz corpus, as the current mBERT results suggest that a language-specific encoder could push F1 beyond 0.93 while eliminating the dependency on a model trained across 104 languages. Additional planned work includes finite-state transducer-based suffix segmentation, a three-class Real/Fake/Satire taxonomy to address satirical content misclassification, cross-lingual

transfer from Turkish, and claims-based verification via Kyrgyz entity knowledge graphs.

REFERENCES

- [1] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, “Fake news detection on social media: A data mining perspective,” *ACM SIGKDD Explor. Newsl.*, vol. 19, no. 1, pp. 22–36, 2017.
- [2] Z. Khanam, B. N. Alwasel, H. Sirafi, and M. Rashid, “Fake news detection using machine learning approaches,” *IOP Conf. Ser.: Mater. Sci. Eng.*, vol. 1099, p. 012040, 2021.
- [3] N. N. Prachi, M. Habibullah, M. E. H. Rafi, E. Alam, and R. Khan, “Detection of fake news using machine learning and natural language processing algorithms,” *J. Adv. Inf. Technol.*, vol. 13, no. 6, pp. 652–661, 2022.
- [4] F. A. Alshuwaier and F. A. Alsulaiman, “Fake news detection using machine learning and deep learning algorithms: A comprehensive review and future perspectives,” *Computers*, vol. 14, p. 394, 2025.
- [5] F. Gereme, W. Zhu, T. Ayall, and D. Alemu, “Combating fake news in low-resource languages: Amharic fake news detection accompanied by resource crafting,” *Information*, vol. 12, p. 20, 2021.
- [6] E. M. Nagoudi, A. Elmadany, M. Abdul-Mageed, T. Alhindi, and H. Cavusoglu, “Machine generation and detection of Arabic manipulated and fake news,” in *Proc. 4th Workshop Arabic NLP, ACL*, pp. 1–12, 2020.
- [7] M. Subramanian, A. Sathya Narayanan, and S. Rajasekar, “Fake news detection in the Malayalam language using XLM-RoBERTa,” in *Proc. Int. Conf. Comput. Commun. Inform. (ICCCI)*, 2023.
- [8] D. Mouratidis, M. Nikiforos, and K. Kermanidis, “A survey on explainable AI for fake news detection with NLP,” in *Proc. 13th Hellenic Conf. AI (SETN)*, 2024.
- [9] A. Torunoglu-Selamet and G. Eryigit, “A cascaded approach for social media text normalization of Turkish,” in *Proc. Workshop Noisy User-Generated Text, ACL*, pp. 62–70, 2016.
- [10] P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov, “Enriching word vectors with subword information,” *Trans. Assoc. Comput. Linguist.*, vol. 5, pp. 135–146, 2017.
- [11] A. Yeshpanov, Y. Khassanov, and H. A. Varol, “KazNERD: Kazakh named entity recognition dataset,” in *Proc. LREC*, 2022.
- [12] L. Ivancova, M. Pikuliak, and M. Simko, “Slovak fake news detection using morphological and semantic features,” in *Proc. TSD*, Springer, pp. 312–322, 2022.
- [13] L. Johanson and E. A. Csató, Eds., *The Turkic Languages*. London: Routledge, 1998, ch. 4: Evidentiality and epistemic modality, pp. 197–211.
- [14] M. Ott, Y. Choi, C. Cardie, and J. T. Hancock, “Finding deceptive opinion spam by any stretch of the imagination,” in *Proc. ACL*, pp. 309–319, 2011.
- [15] W. Y. Wang, “‘Liar, Liar Pants on Fire’: A new benchmark dataset for fake news detection,” in *Proc. ACL*, pp. 422–426, 2017.
- [16] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, pp. 4171–4186, 2019.
- [17] A. Conneau *et al.*, “Unsupervised cross-lingual representation learning at scale,” in *Proc. ACL*, pp. 8440–8451, 2020.
- [18] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.
- [19] I. Mashrapov, “Kyrgyz Fake News Detection: MorphAware Fusion Codebase,” GitHub, 2025. [Online]. Available: <https://github.com/Imanbek-Mashrapov/Fake-News-Detection>

APPENDIX

The following checklist confirms reproducibility of all reported results:

- **Dataset:** `kyrgyz_news_final_unified.csv` released at GitHub repository [19] (1,683 articles)
- **Codebase:** `pipeline.py` (767 lines, zero external ML dependencies) released at [19]
- **Random seeds:** 42, 123, 777 for all experiments; mean values reported

Algorithm 1 MorphAware Fusion Training Procedure

Require: Training texts T , labels y , hyperparams $(\eta, \text{epochs}, \alpha)$

Ensure: Trained weight vectors $\mathbf{W}_1, \mathbf{W}_2$, bias b

```
1: Preprocess each  $t \in T$  (Unicode norm, URL removal)
2: Fit TF-IDF on  $T \rightarrow$  vocabulary  $V$  ( $|V| = 5000$ )
3: for each  $t \in T$  do
4:    $\mathbf{H}_{\text{tfidf}}(t) \leftarrow \text{TF-IDF transform}(t)$ 
5:    $\mathbf{H}_{\text{morph}}(t) \leftarrow \text{extract\_morph\_features}(t)$ 
6: end for
7:  $\mathbf{W}_1 \sim \text{Xavier\_uniform}(|V|)$ 
8:  $\mathbf{W}_2 \sim \text{Uniform}(-\alpha/\sqrt{15}, +\alpha/\sqrt{15}), \alpha = 2.0$ 
9:  $b \leftarrow 0$ 
10: for epoch = 1 to epochs do
11:   for each  $(\mathbf{H}_{\text{tfidf}_i}, \mathbf{H}_{\text{morph}_i}, y_i)$  do
12:      $\hat{y}_i = \sigma(\mathbf{W}_1 \cdot \mathbf{H}_{\text{tfidf}_i} + \mathbf{W}_2 \cdot \mathbf{H}_{\text{morph}_i} + b)$ 
13:      $\mathbf{W}_1 \leftarrow \mathbf{W}_1 - \eta \cdot (\hat{y}_i - y_i) \cdot \mathbf{H}_{\text{tfidf}_i}$ 
14:      $\mathbf{W}_2 \leftarrow \mathbf{W}_2 - \eta \cdot (\hat{y}_i - y_i) \cdot \mathbf{H}_{\text{morph}_i}$ 
15:      $b \leftarrow b - \eta \cdot (\hat{y}_i - y_i)$ 
16:   end for
17:    $\eta \leftarrow \eta_0 \cdot 0.5 \cdot (1 + \cos(\pi \cdot \text{epoch}/\text{epochs}))$ 
18: end for
19: return  $\mathbf{W}_1, \mathbf{W}_2, b$ 
```

- **Temporal split:** February 2025 (train $\leq$ Jan 2025, test $\geq$ Feb 2025)
- **TF-IDF:** max_features = 5,000, min_df = 2, smooth_idf = True
- **$\mathbf{H}_{\text{morph}}$:** 15-dimensional vector; suffix lexicons in pipeline.py lines 28–58
- **MorphAware Fusion:** $\eta = 0.05$, epochs = 60, $\alpha = 2.0$
- **mBERT:** bert-base-multilingual-cased via HuggingFace
- **SHAP:** $\geq 0.41.0$; TreeExplainer for LR-based model
- **Runtime:** full suite < 3 minutes on CPU (Intel Core i5, 8 GB RAM)